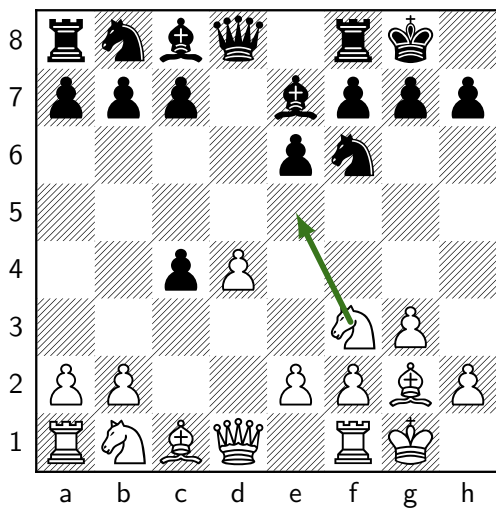


Melanie-ANALOGY — Catalan

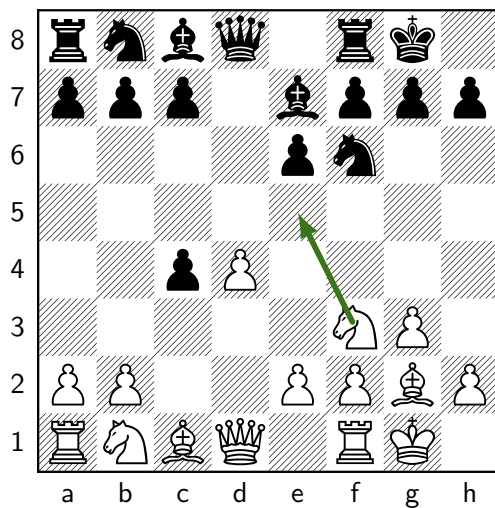
Melanie (Deep_Vision)

Abstract

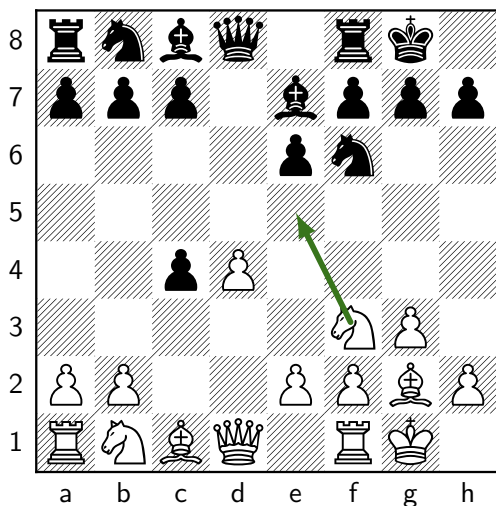
Every move below was produced by relational analogy to named master games (human and/or top-engine – the exact source is named per move). Each figure shows our position with Melanie's move (a) and the top retrieved analog positions (b–d) with the master move played there; the caption names the exact source and the piece-role correspondence under μ – cross-type maps (a rook playing a queen's role) are called out, as no material/eval engine explains a move that way.



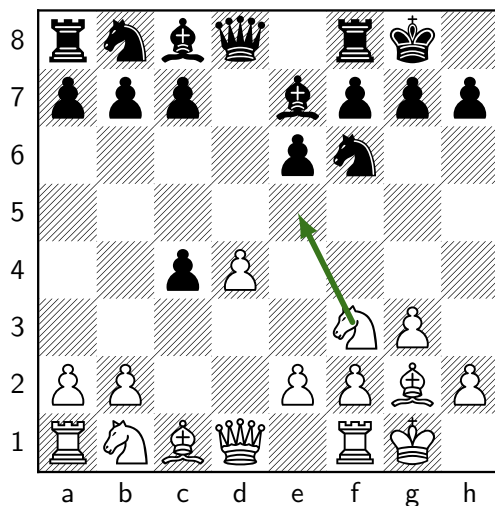
(a)



(b)

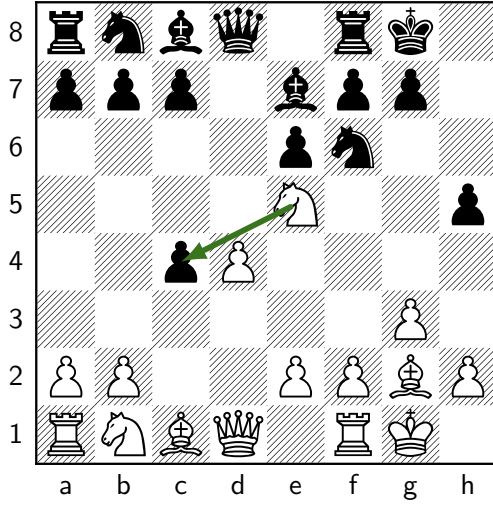


(c)

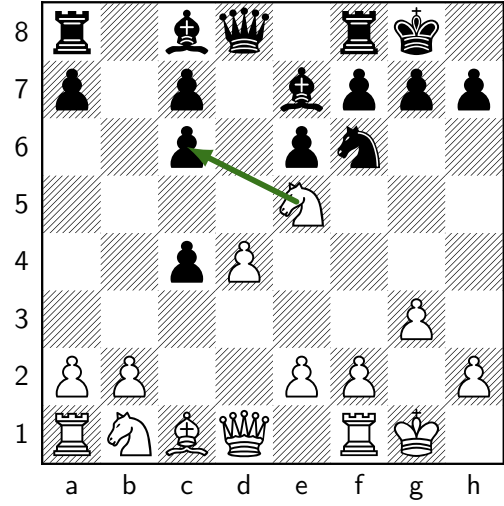


(d)

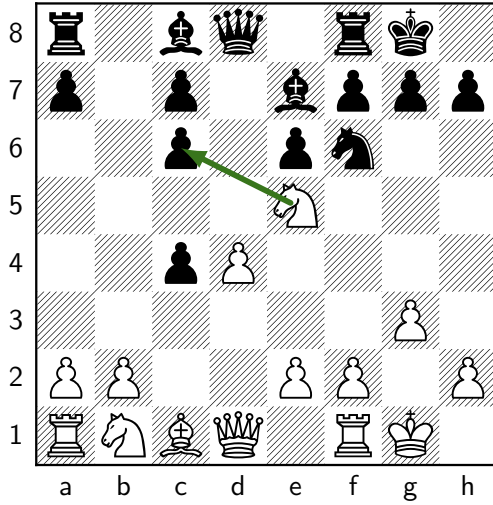
Figure 1: **Move 7, White: Ne5.** (a) Our position; Melanie plays Ne5 (f3e5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#3464` (ply 12) [structural distance $d=0$, QAP obj=74]: the master played Ne5; under μ our *Nf3* plays the role of that game’s *Nf3*; cross-type roles: our *Pa2* as their *Bc1*, our *Ph2* as their *Bg2*; transferred move f3e5 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#3465` (ply 12) [structural distance $d=0$, QAP obj=74]: the master played Ne5; under μ our *Nf3* plays the role of that game’s *Nf3*; cross-type roles: our *Pa2* as their *Bc1*, our *Ph2* as their *Bg2*; transferred move f3e5 (**endorses this move**). (d) Retrieved from `ccrl_top_p0001#3065` (ply 12) [structural distance $d=0$, QAP obj=74]: the master played Ne5; under μ our *Nf3* plays the role of that game’s *Nf3*; cross-type roles: our *Pa2* as their *Bc1*, our *Ph2* as their *Bg2*; transferred move f3e5 (**endorses this move**). Synthesis: 14 retrieved analogs transferred to Ne5 (summed QAP objective 928, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



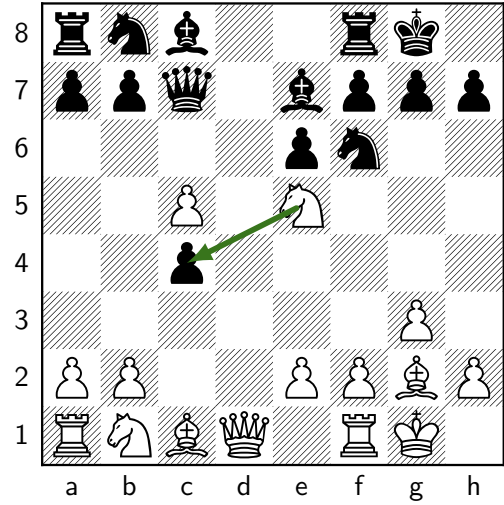
(a)



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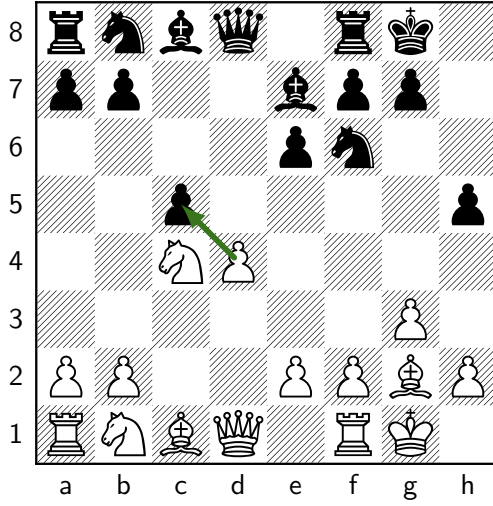


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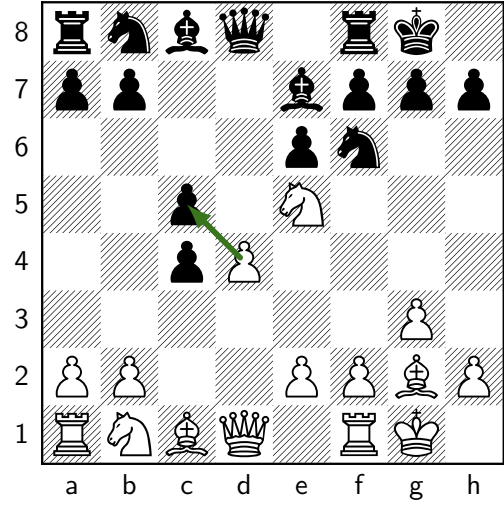


(d)

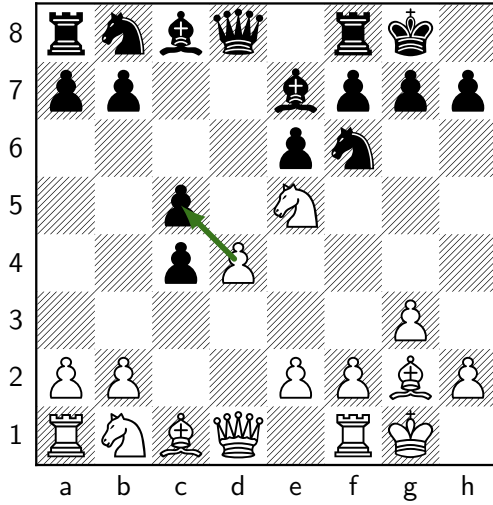
Figure 2: **Move 8, White: Nxc4**. (a) Our position; Melanie plays Nxc4 (e5c4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0000#3464 (ply 16) [structural distance $d=37$, QAP obj=140]: the master played Nxc6; under μ our Ne5 plays the role of that game’s Ne5; cross-type roles: our pb7 as their Ra1, our Ra1 as their pa7; transferred move e5c4 (**endorses this move**). (c) Retrieved from ccr1_top_p0000#3465 (ply 16) [structural distance $d=37$, QAP obj=140]: the master played Nxc6; under μ our Ne5 plays the role of that game’s Ne5; cross-type roles: our pb7 as their Ra1, our Ra1 as their pa7; transferred move e5c4 (**endorses this move**). (d) Retrieved from ccr1_top_p0002#2426 (ply 16) [structural distance $d=44$, QAP obj=47]: the master played Nxc4; under μ our Ne5 plays the role of that game’s Ne5; cross-type roles: our Bc1 as their Pa2, our Ne5 as their qc7; transferred move e5c4 (**endorses this move**). Synthesis: 10 retrieved analogs transferred to Nxc4 (summed QAP objective 533, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



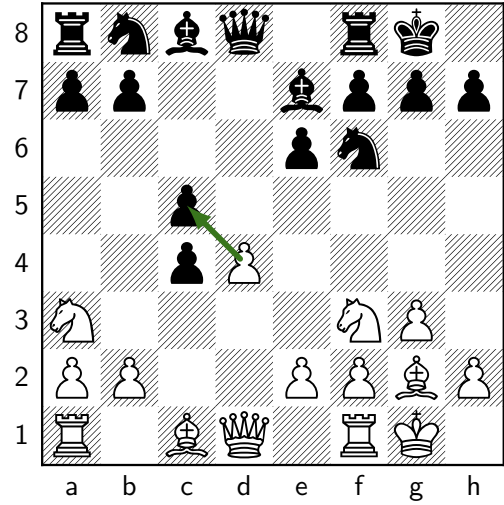
(a)



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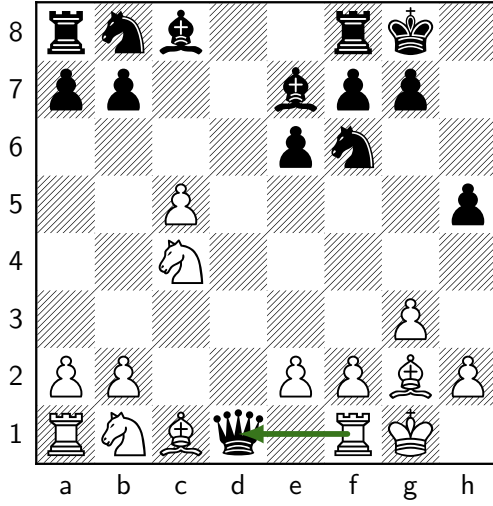


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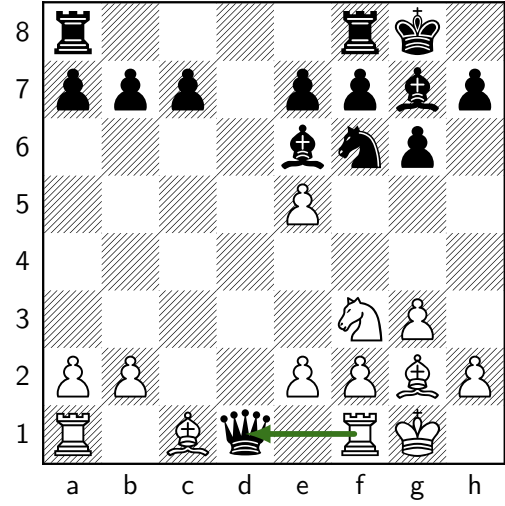


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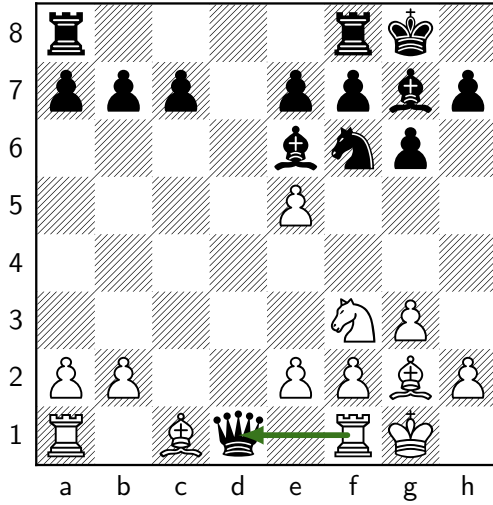
Figure 3: **Move 9, White: dxc5.** (a) Our position; Melanie plays dxc5 (d4c5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0002#2426 (ply 14) [structural distance $d=51$, QAP obj=148]: the master played dxc5; under μ our $Pd4$ plays the role of that game’s $Pd4$; cross-type roles: our $Nc4$ as their $Ra1$, our $Pa2$ as their $Bc1$; transferred move d4c5 (**endorses this move**). (c) Retrieved from ccr1_top_p0002#2427 (ply 14) [structural distance $d=51$, QAP obj=148]: the master played dxc5; under μ our $Pd4$ plays the role of that game’s $Pd4$; cross-type roles: our $Nc4$ as their $Ra1$, our $Pa2$ as their $Bc1$; transferred move d4c5 (**endorses this move**). (d) Retrieved from ccr1_top_p0000#2982 (ply 14) [structural distance $d=58$, QAP obj=121]: the master played dxc5; under μ our $Pd4$ plays the role of that game’s $Pd4$; cross-type roles: our $Pg3$ as their $Ra1$, our $Nb1$ as their $Pa2$; transferred move d4c5 (**endorses this move**). Synthesis: 20 retrieved analogs transferred to dxc5 (summed QAP objective 1805, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



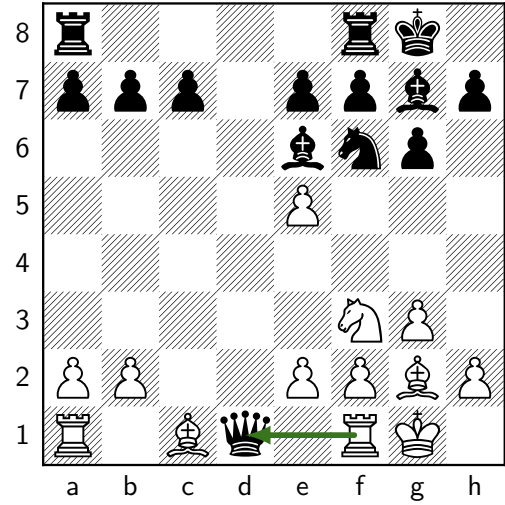
(a)



(b)

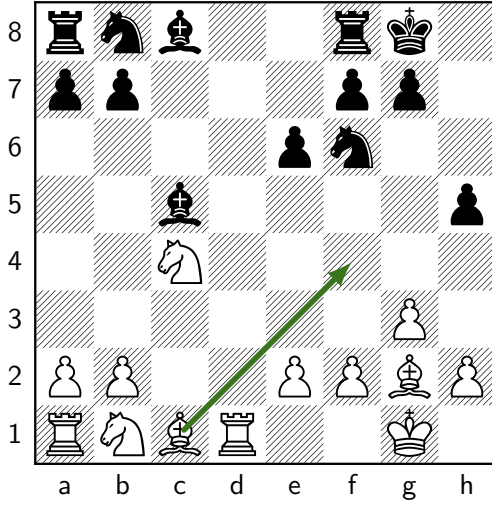


(c)

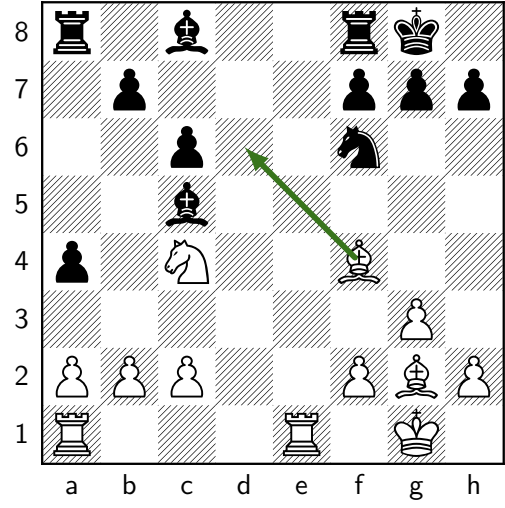


(d)

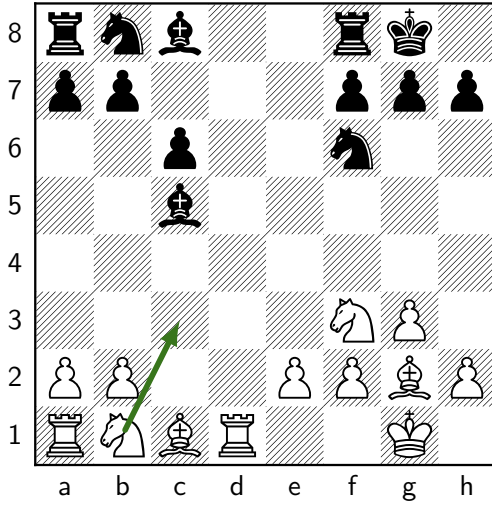
Figure 4: **Move 10, White: Rxd1.** (a) Our position; Melanie plays Rxd1 (f1d1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#2008` (ply 20) [structural distance $d=78$, QAP obj=124]: the master played Rxd1; under μ our *Rf1* plays the role of that game’s *Rf1*; cross-type roles: our *Rf1* as their *Bc1*, our *qd1* as their *Pe5*; transferred move f1d1 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#1562` (ply 20) [structural distance $d=78$, QAP obj=124]: the master played Rxd1; under μ our *Rf1* plays the role of that game’s *Rf1*; cross-type roles: our *Rf1* as their *Bc1*, our *qd1* as their *Pe5*; transferred move f1d1 (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#2975` (ply 20) [structural distance $d=78$, QAP obj=124]: the master played Rxd1; under μ our *Rf1* plays the role of that game’s *Rf1*; cross-type roles: our *Rf1* as their *Bc1*, our *qd1* as their *Pe5*; transferred move f1d1 (**endorses this move**). Synthesis: 41 retrieved analogs transferred to Rxd1 (summed QAP objective 4380, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



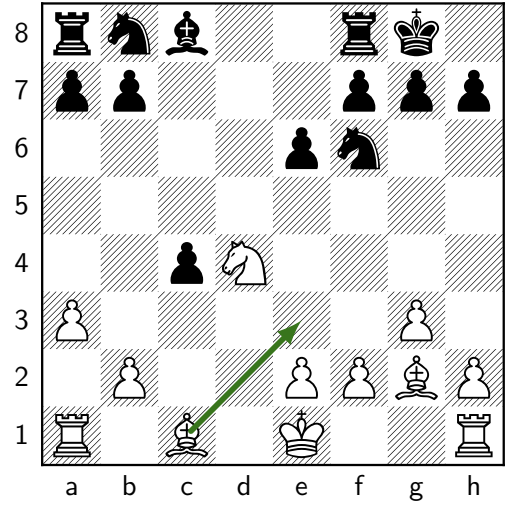
(a)



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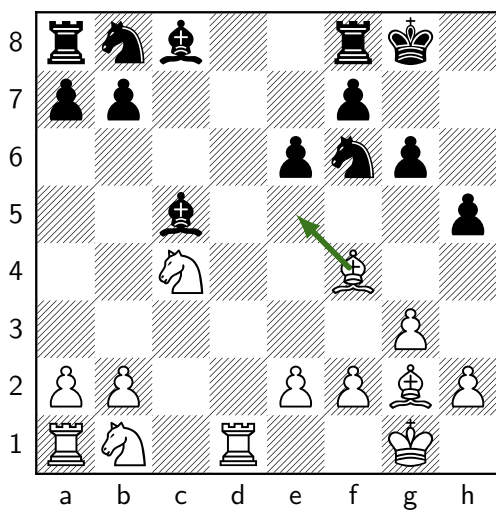


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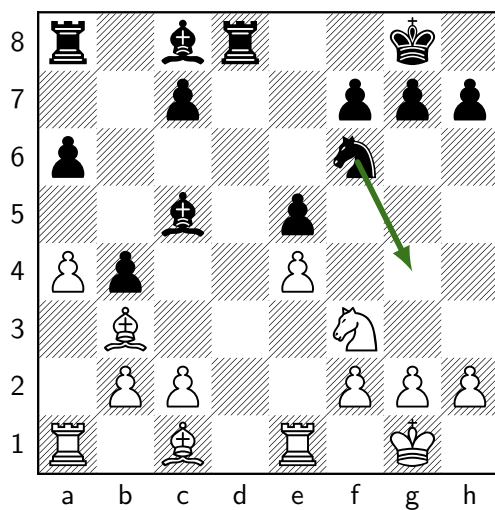


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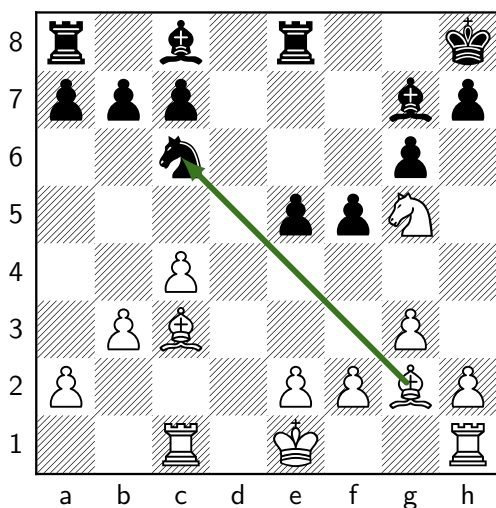
Figure 5: **Move 11, White: Bf4**. (a) Our position; Melanie plays Bf4 (c1f4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0003#648 (ply 36) [structural distance $d=51$, QAP obj=75]: the master played Bd6; under μ our Bc1 plays the role of that game’s Bf4; cross-type roles: our Nb1 as their Pc2, our bc8 as their pb7; transferred move c1f4 (**endorses this move**). (c) Retrieved from ccr1_top_p0003#2697 (ply 20) [structural distance $d=37$, QAP obj=61]: the master played Nc3; under μ our Bc1 plays the role of that game’s Nb1 (a **cross-type** correspondence); cross-type roles: our Pe2 as their Ra1, our pb7 as their Bc1; transferred move c1f4 (**endorses this move**). (d) Retrieved from ccr1_top_p0001#1378 (ply 20) [structural distance $d=55$, QAP obj=44]: the master played Be3; under μ our Bc1 plays the role of that game’s Bc1; cross-type roles: our Nc4 as their Pf2, our Nb1 as their Ph2; transferred move c1f4 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Bf4 (summed QAP objective 248, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



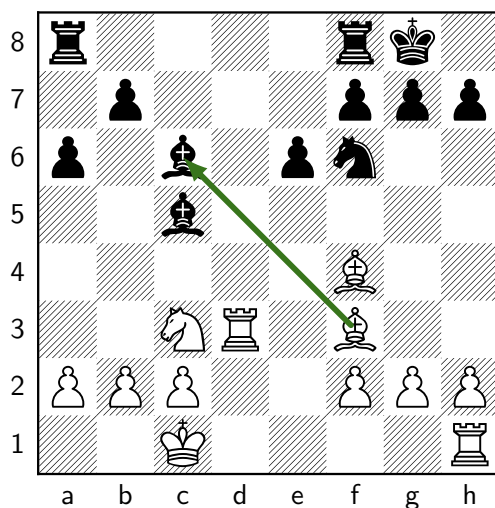
(a)



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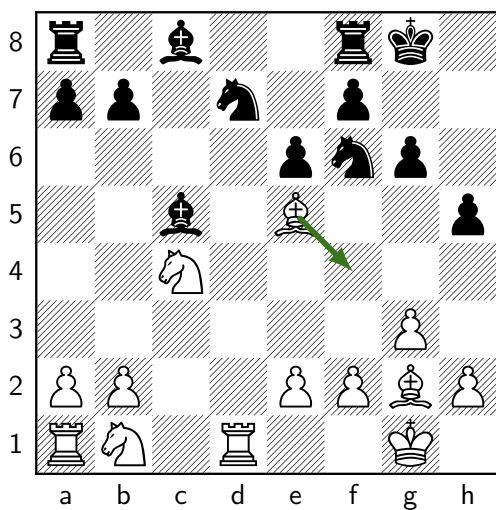


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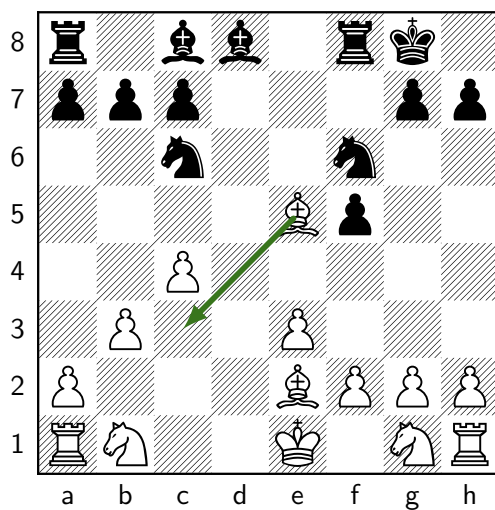


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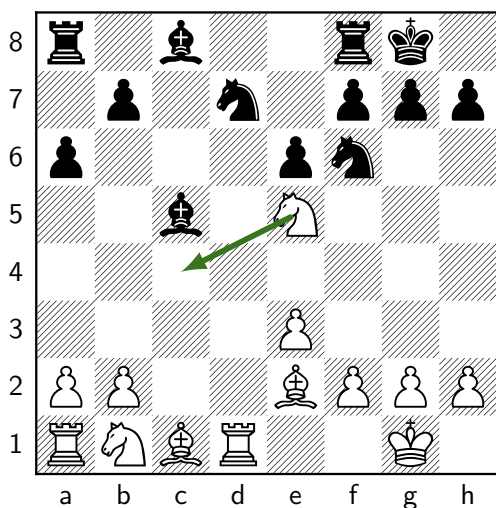
Figure 6: **Move 12, White: Be5.** (a) Our position; Melanie plays Be5 (f4e5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1188` (ply 27) [structural distance $d=60$, QAP obj=62]: the master played Ng4; under μ our Bf4 plays the role of that game’s Nf3 (a **cross-type** correspondence); cross-type roles: our Ph2 as their Ra8, our Ra1 as their Bc8; transferred move f4e5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3104` (ply 28) [structural distance $d=65$, QAP obj=47]: the master played Bxc6; under μ our Bf4 plays the role of that game’s Bg2; cross-type roles: our Pb2 as their Rh1, our bc8 as their Pe2; transferred move f4e5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#1257` (ply 30) [structural distance $d=66$, QAP obj=31]: the master played Bxc6; under μ our Bf4 plays the role of that game’s Bf3; cross-type roles: our nf6 as their Rh1, our Nb1 as their Pb2; transferred move f4e5 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Be5 (summed QAP objective 228, the strongest correspondence among the candidate moves); Melanie committed it via the search selector.



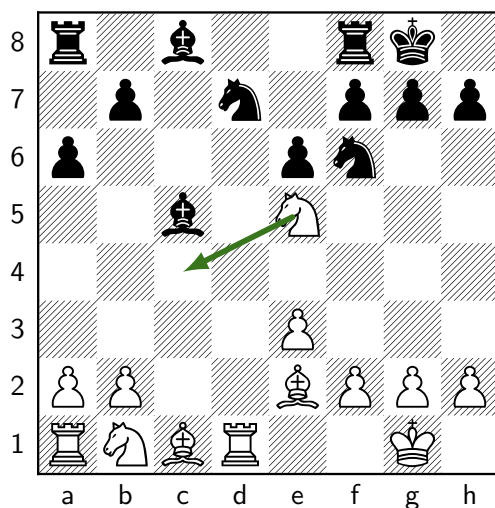
(a)



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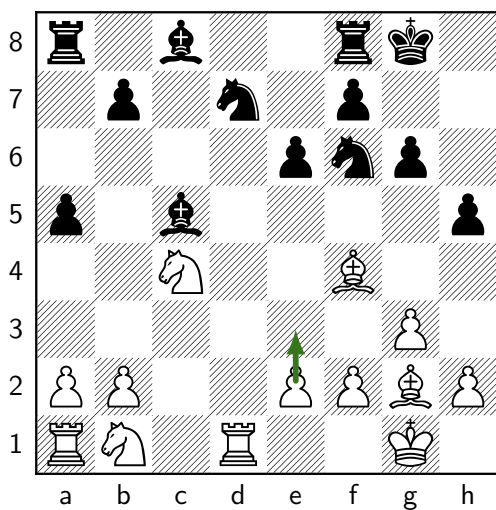


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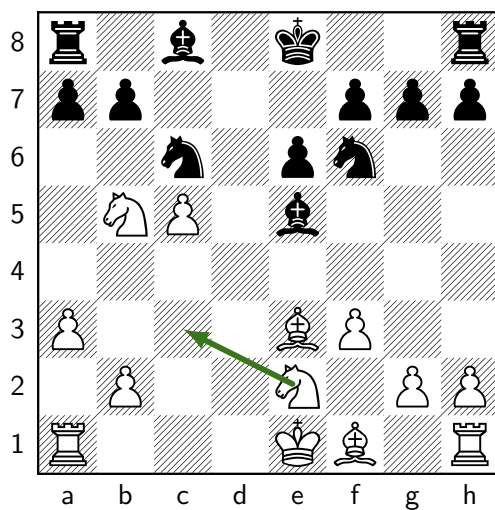


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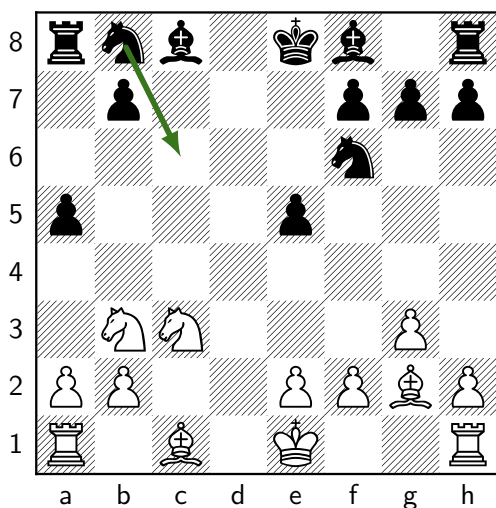
Figure 7: **Move 13, White: Bf4**. (a) Our position; Melanie plays Bf4 (e5f4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0001#471 (ply 18) [structural distance $d=72$, QAP obj=54]: the master played Bc3; under μ our Be5 plays the role of that game’s Be5; cross-type roles: our pb7 as their Ng1, our bc5 as their Rh1; transferred move e5f4 (**endorses this move**). (c) Retrieved from ccr1_top_p0003#2118 (ply 22) [structural distance $d=46$, QAP obj=27]: the master played Nc4; under μ our Be5 plays the role of that game’s Ne5 (a **cross-type** correspondence); cross-type roles: our bc5 as their Ra1, our Pb2 as their Nb1; transferred move e5f4 (**endorses this move**). (d) Retrieved from ccr1_top_p0002#1326 (ply 20) [structural distance $d=46$, QAP obj=27]: the master played Nc4; under μ our Be5 plays the role of that game’s Ne5 (a **cross-type** correspondence); cross-type roles: our bc5 as their Ra1, our Pb2 as their Nb1; transferred move e5f4 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Bf4 (summed QAP objective 136, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



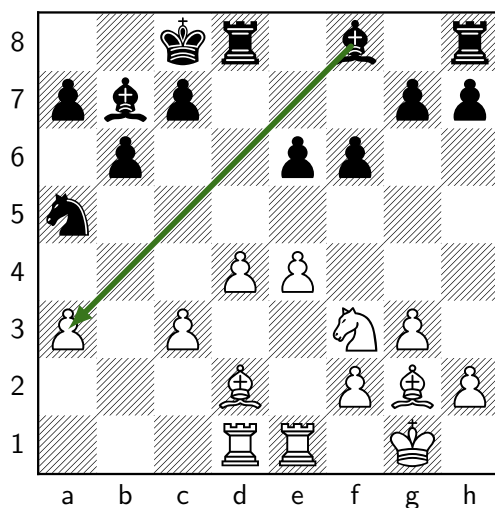
(a)



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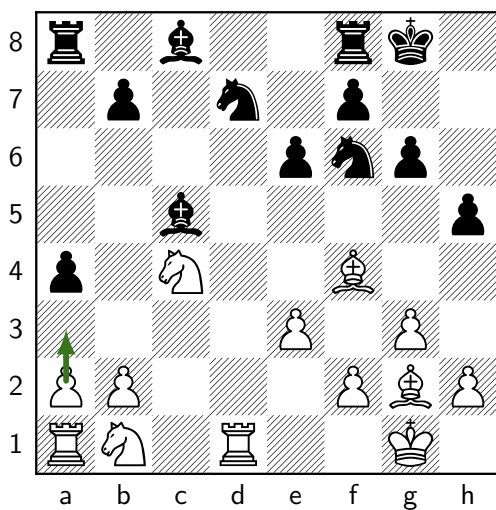


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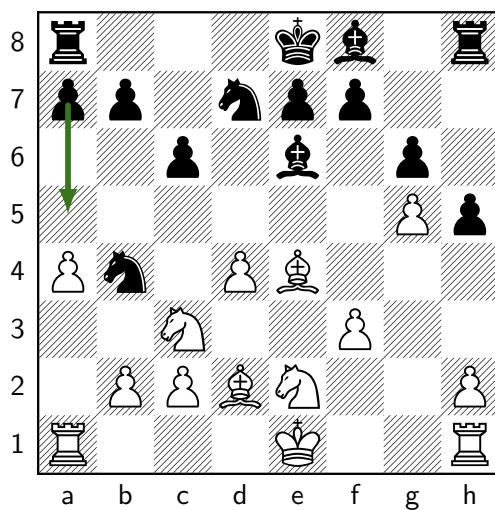


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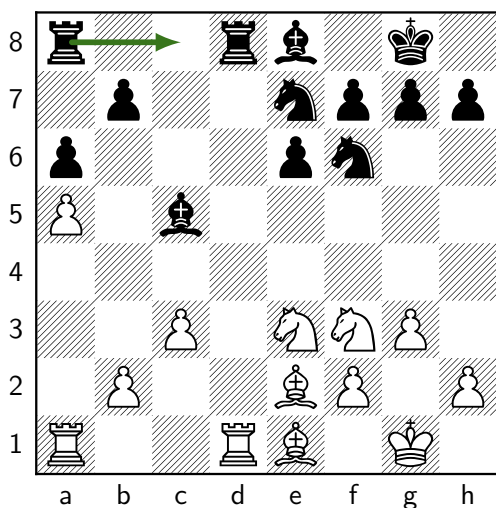
Figure 8: **Move 14, White: e3**. (a) Our position; Melanie plays **e3** (**e2e3**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#3968` (ply 28) [structural distance $d=72$, QAP obj=70]: the master played **Nec3**; under μ our *Pe2* plays the role of that game’s *Ne2* (a **cross-type** correspondence); cross-type roles: our *Ph2* as their *Bf1*, our *Pa2* as their *Rh1*; transferred move **e2e3** (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#1066` (ply 21) [structural distance $d=68$, QAP obj=45]: the master played **Nc6**; under μ our *Pe2* plays the role of that game’s *Nb1* (a **cross-type** correspondence); cross-type roles: our *nd7* as their *Bc8*, our *nf6* as their *Bf8*; transferred move **e2e3** (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#1878` (ply 29) [structural distance $d=79$, QAP obj=65]: the master played **Bxa3**; under μ our *Nc4* plays the role of that game’s *Bf1* (a **cross-type** correspondence); cross-type roles: our *Bg2* as their *Rd8*, our *Nb1* as their *Rh8*; transferred move **c4a5**. Synthesis: 2 retrieved analogs transferred to **e3** (summed QAP objective 115, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



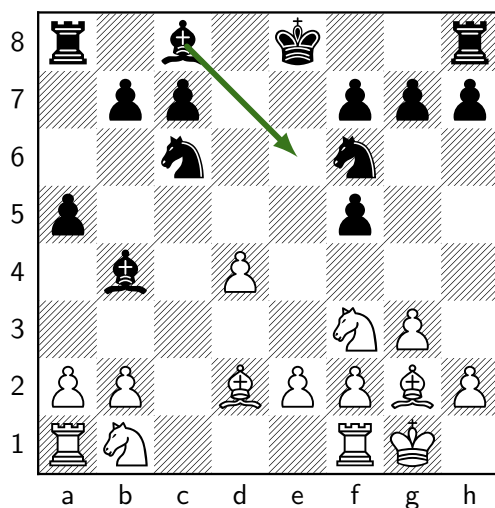
(a)



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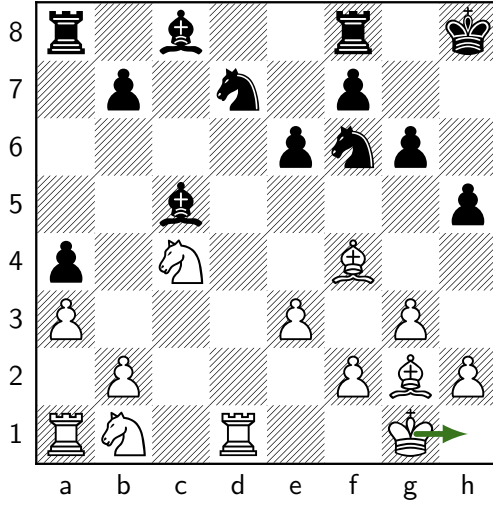


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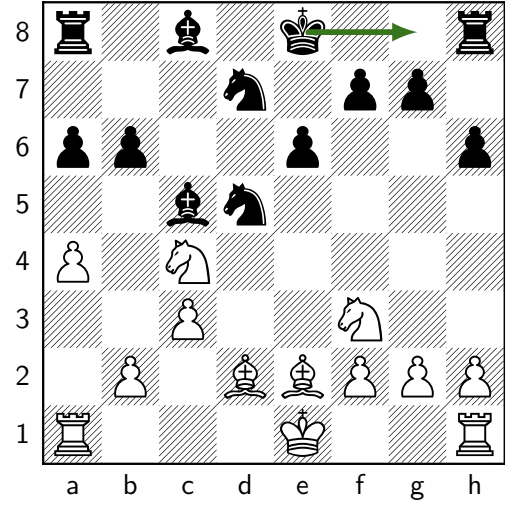


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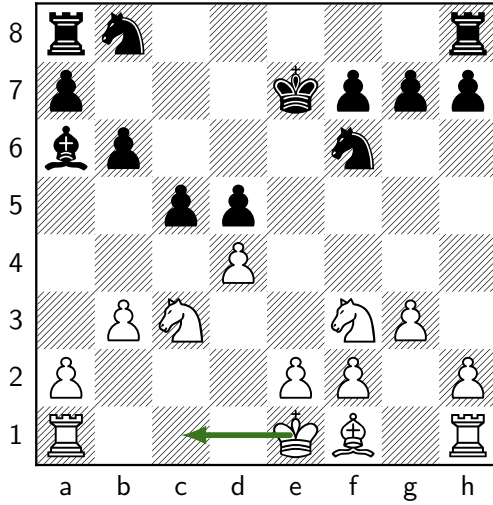
Figure 9: **Move 15, White: a3.** (a) Our position; Melanie plays a3 (a2a3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#368` (ply 31) [structural distance $d=74$, QAP obj=55]: the master played a5; under μ our *Pa2* plays the role of that game’s *Pa2*; cross-type roles: our *nf6* as their *Ra8*, our *Nc4* as their *Bf8*; transferred move a2a3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#2154` (ply 37) [structural distance $d=74$, QAP obj=19]: the master played Rac8; under μ our *Pa2* plays the role of that game’s *Ra1* (a **cross-type** correspondence); cross-type roles: our *Pe3* as their *Rd8*, our *Pb2* as their *Be8*; transferred move a2a3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#1233` (ply 21) [structural distance $d=72$, QAP obj=15]: the master played Be6; under μ our *Pa2* plays the role of that game’s *Bc1* (a **cross-type** correspondence); cross-type roles: our *Bg2* as their *Ra8*, our *Pb2* as their *Rh8*; transferred move a2a3 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to a3 (summed QAP objective 108, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



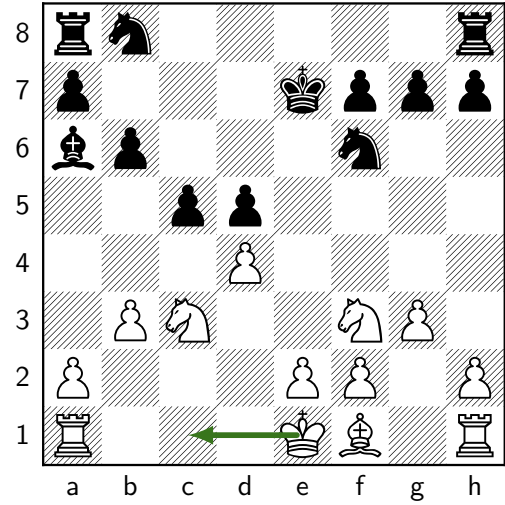
(a)



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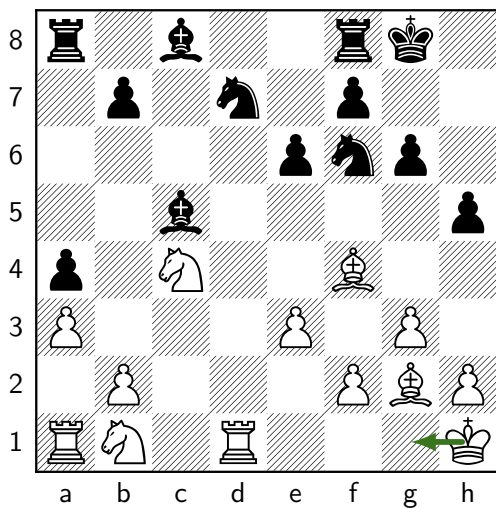


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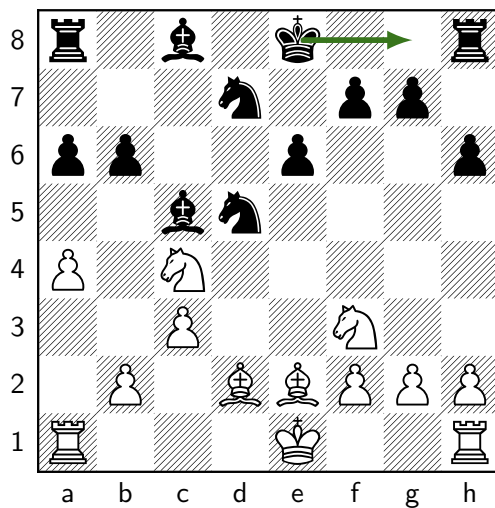


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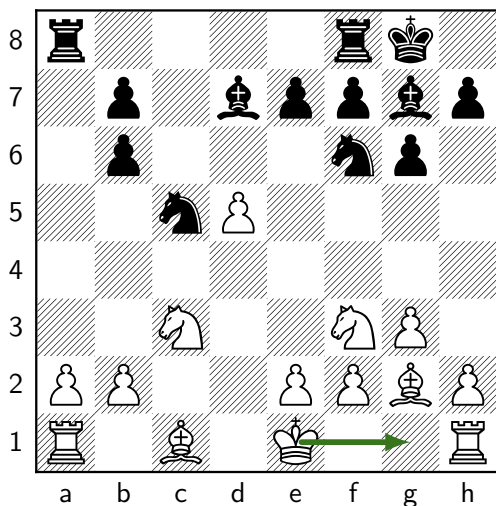
Figure 10: **Move 16, White: Kh1.** (a) Our position; Melanie plays Kh1 (g1h1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2216` (ply 25) [structural distance $d=107$, QAP obj=52]: the master played 0-0; under μ our *Kg1* plays the role of that game’s *Ke1*; cross-type roles: our *Pe3* as their *Ra8*, our *ra8* as their *Bc8*; transferred move g1h1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3189` (ply 22) [structural distance $d=107$, QAP obj=39]: the master played 0-0-0; under μ our *Kg1* plays the role of that game’s *Ke1*; cross-type roles: our *Nb1* as their *Rh1*, our *Ra1* as their *Pa2*; transferred move g1h1 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#3190` (ply 22) [structural distance $d=107$, QAP obj=39]: the master played 0-0-0; under μ our *Kg1* plays the role of that game’s *Ke1*; cross-type roles: our *Nb1* as their *Rh1*, our *Ra1* as their *Pa2*; transferred move g1h1 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Kh1 (summed QAP objective 168, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



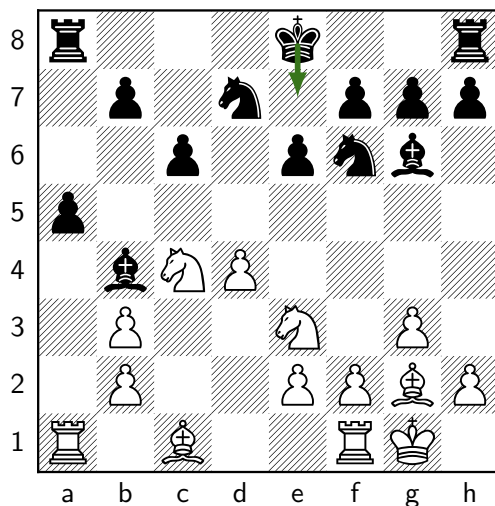
(a)



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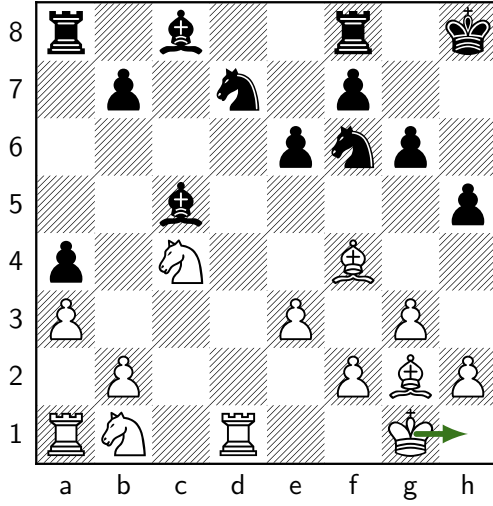


(c)

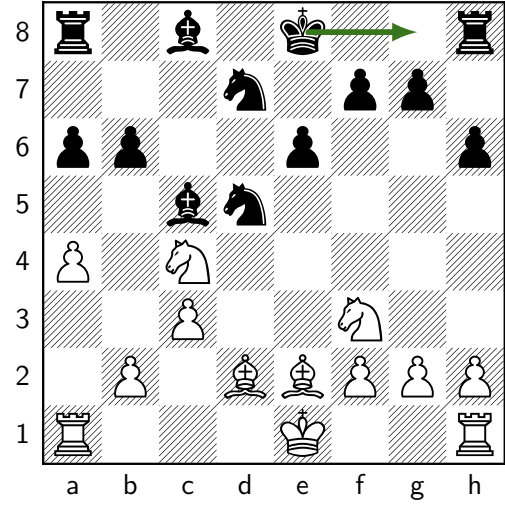


(d)

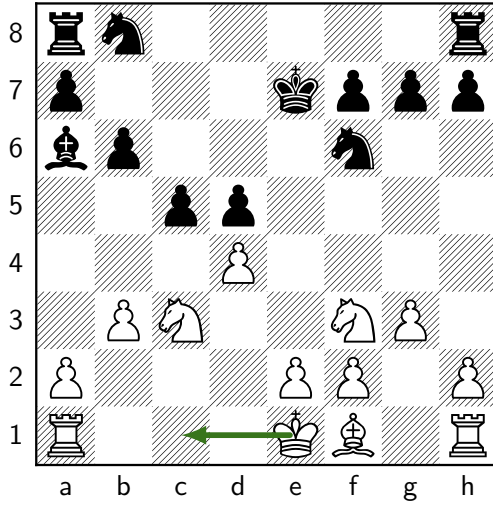
Figure 11: **Move 17, White: Kg1**. (a) Our position; Melanie plays Kg1 (**h1g1**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2216` (ply 25) [structural distance $d=85$, QAP obj=58]: the master played 0-0; under μ our *Kh1* plays the role of that game's *Ke1*; cross-type roles: our *Ph2* as their *Ra8*, our *pg6* as their *Bc8*; transferred move **h1g1 (endorses this move)**. (c) Retrieved from `ccr1_top_p0001#2773` (ply 22) [structural distance $d=79$, QAP obj=46]: the master played 0-0; under μ our *Kh1* plays the role of that game's *Ke1*; cross-type roles: our *pe6* as their *Ra1*, our *pf7* as their *Bc1*; transferred move **h1g1 (endorses this move)**. (d) Retrieved from `ccr1_top_p0000#3287` (ply 29) [structural distance $d=81$, QAP obj=40]: the master played *Ke7*; under μ our *Kh1* plays the role of that game's *Ke1*; cross-type roles: our *pa4* as their *Rh8*, our *Rd1* as their *Pb7*; transferred move **h1g1 (endorses this move)**. Synthesis: 4 retrieved analogs transferred to Kg1 (summed QAP objective 166, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



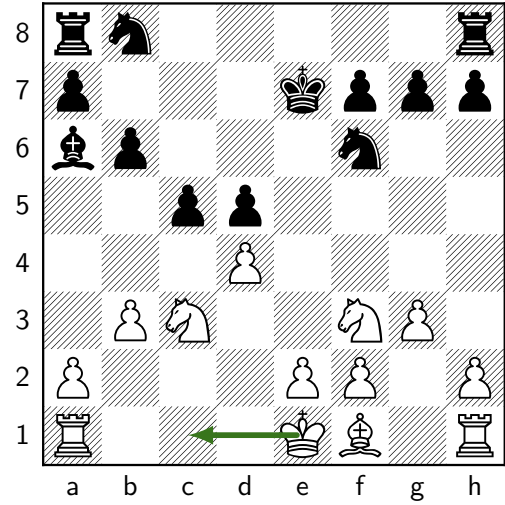
(a)



(b)



(c)



(d)

Figure 12: **Move 18, White: Kh1.** (a) Our position; Melanie plays Kh1 (g1h1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0000#2216 (ply 25) [structural distance $d=107$, QAP obj=52]: the master played 0-0; under μ our Kg1 plays the role of that game’s Ke1; cross-type roles: our Pe3 as their Ra8, our ra8 as their Bc8; transferred move g1h1 (**endorses this move**). (c) Retrieved from ccr1_top_p0001#3189 (ply 22) [structural distance $d=107$, QAP obj=39]: the master played 0-0-0; under μ our Kg1 plays the role of that game’s Ke1; cross-type roles: our Nb1 as their Rh1, our Ra1 as their Pa2; transferred move g1h1 (**endorses this move**). (d) Retrieved from ccr1_top_p0001#3190 (ply 22) [structural distance $d=107$, QAP obj=39]: the master played 0-0-0; under μ our Kg1 plays the role of that game’s Ke1; cross-type roles: our Nb1 as their Rh1, our Ra1 as their Pa2; transferred move g1h1 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Kh1 (summed QAP objective 168, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.